

TrendScope Architectures

Technical Flowcharts & Visual Architecture
Diagrams

TrendScope Team Onboarding & Handover package

Creator Analytics & YouTube Trends Explorer Platform

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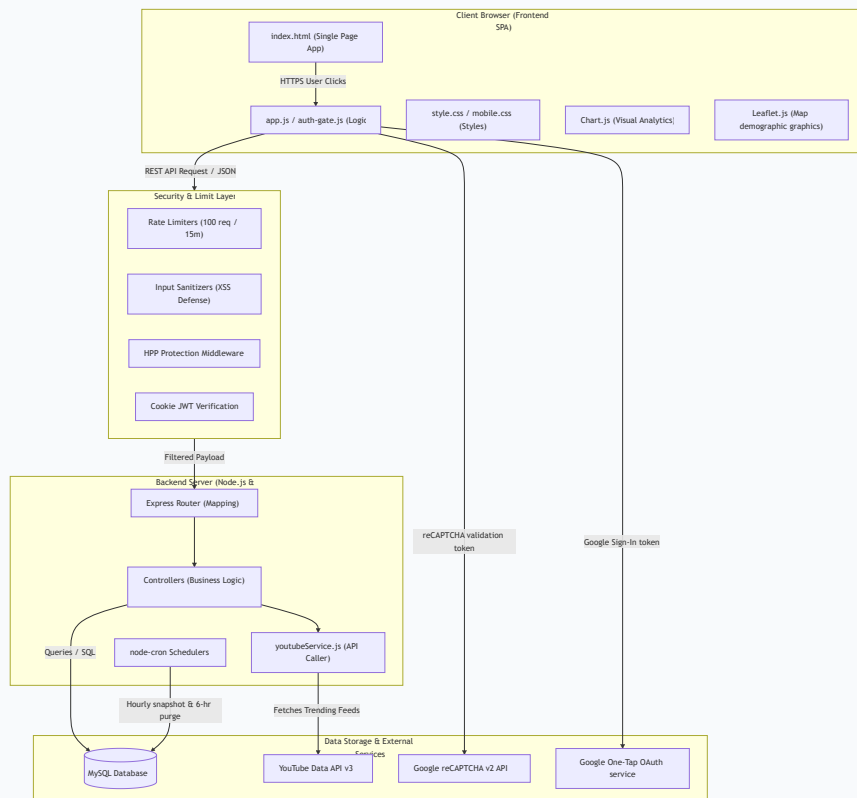
TrendScope — System Architecture & Data Flow Diagrams

This document explains the technical architecture, operational pipelines, and data flows of the **TrendScope** system using visual Mermaid diagrams. It is designed to help you quickly understand how frontend, backend, database, and third-party APIs work together!

1. High-Level System Architecture

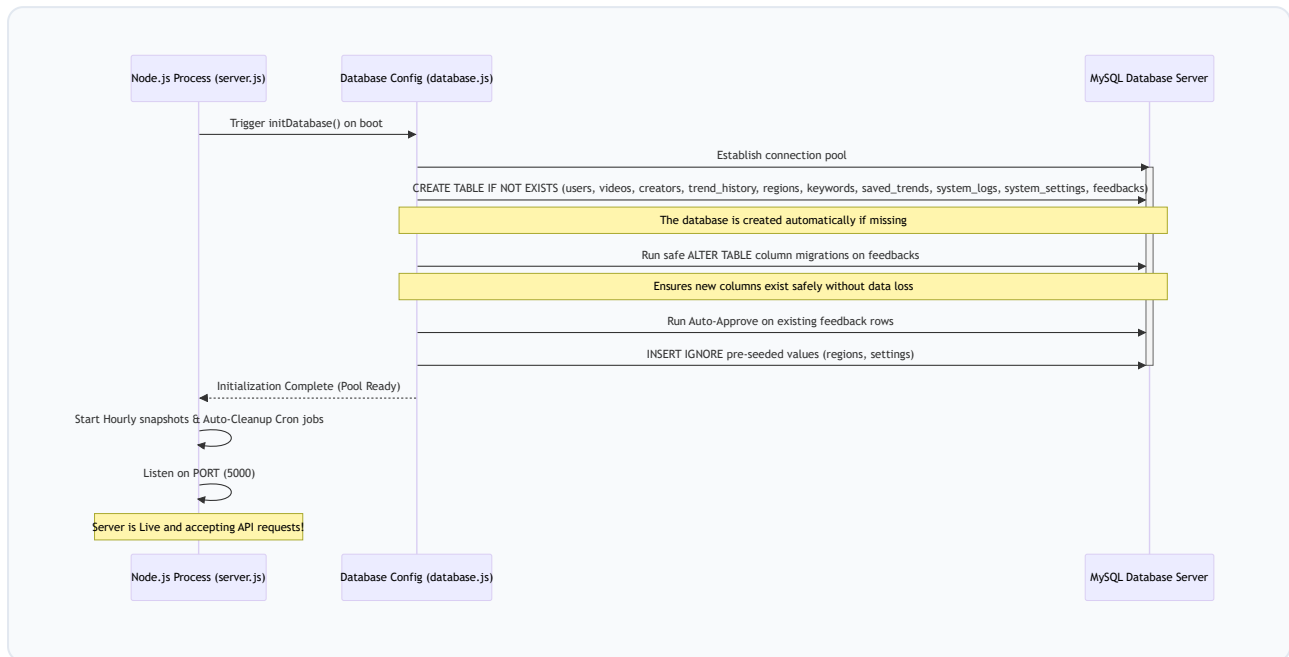
TrendScope uses a classic **Client-Server-Database** architecture.

- The **Frontend SPA** (Single Page Application) runs entirely in the user's web browser.
- The **Backend Express Server** handles the business logic, session checks, security limits, and schedules.
- The **MySQL Database** stores persistent records (users, history, logs) and caches heavy YouTube API feeds.



2. Server Boot & Database Initialization Flow

When the Node.js backend starts up (`npm start`), it goes through a safe, automated initialization pipeline to guarantee database integrity before starting the server.



3. Data Query Flow: Viewing Trends (Caching Strategy)

To save YouTube API quota (which has a daily maximum of 10,000 requests), TrendScope utilizes a highly efficient **Cache-First Caching Strategy**.

 Architecture Diagram

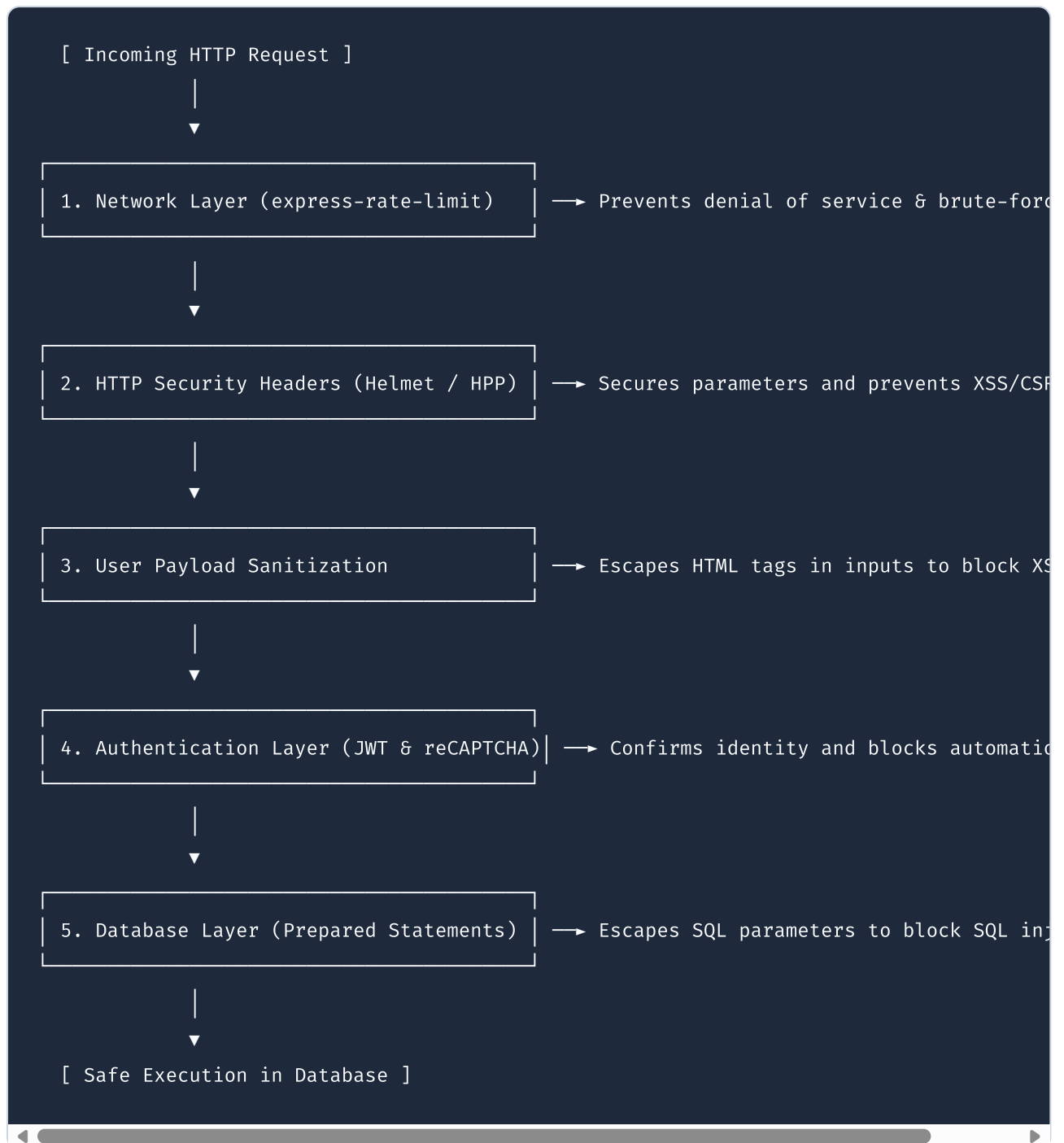
4. Authentication Architecture (JWT & HttpOnly Cookies)

TrendScope implements secure token-based authentication.

 Architecture Diagram

5. Multi-Layer Security Architecture

TrendScope uses a comprehensive **defense-in-depth** strategy to secure every layer of the application.



Happy studying! This visual architecture manual should give you a clear, structural map of the entire TrendScope platform. Let's make something great!